Experiment No. : 7

Aim : Solve real-life problems based on clocks, calendars, weekdays, and repeated events using modular arithmetic, record the remainder-based reasoning used in each case, and explain how cyclic patterns support time management, scheduling, and introductory cryptographic logic.

Requirements : Python, Computer/Laptop, Python IDLE or VS Code, Notebook, Pen, PPT for theory.

Theory :

- **Modular Arithmetic:** It is arithmetic based on remainders. It is useful for problems where values repeat in cycles, such as clocks and weekdays.
- **Clock Arithmetic:** Clock values repeat after 12 hours. Example: 17 mod 12 = 5.
- **Weekday Arithmetic:** Weekdays repeat after 7 days, so the remainder after division by 7 gives the shift in the weekday.
- **Ancient and Modern Use:** Cyclic calculations have applications in traditional calendar and astronomical calculations. Today, modular arithmetic is used in clocks, calendars, scheduling, computing and basic cryptography.

Practice Questions – Solutions

1. **It is 9 o'clock now. What time will it be after 17 hours?**

   Working: 17 mod 12 = 5; 9 + 5 = 14 → 2 o'clock.

   Answer: **2 o'clock**

2. **A movie starts at 4 PM and lasts 29 hours. What time will it end?**

   Working: 29 mod 24 = 5; 4 PM + 5 hours = 9 PM.

   Answer: **9 PM (next day)**

3. **Today is Monday. After 61 days, what day will it be?**

   Working: 61 mod 7 = 5; Monday + 5 days = Saturday.

   Answer: **Saturday**

4. **Today is Friday. After 100 days, what day will it be?**

   Working: 100 mod 7 = 2; Friday + 2 days = Sunday.

   Answer: **Sunday**

5. **A meeting is held every 10 days. The first meeting is on Wednesday. On which weekday will the 9th meetingbe held?**

   Working: (9 − 1) × 10 = 80 days; 80 mod 7 = 3; Wednesday + 3 = Saturday.

   Answer: **Saturday**

6. **A training session is conducted every 15 days. The first session is on Monday. On which weekday will the6th session be held?**

   Working: (6 − 1) × 15 = 75 days; 75 mod 7 = 5; Monday + 5 = Saturday.

   Answer: **Saturday**

7. **A patient takes medicine every 18 hours. The first dose is on Monday at 6:00 AM. What weekday and timewill the 10th dose occur?**

   Working: (10 − 1) × 18 = 162 hours = 6 days 18 hours. Monday 6 AM + 6 days 18 hours = Monday 12 AM.

   Answer: **Monday, 12:00 AM**

8. **A security guard checks a gate every 13 hours. The first check is at 7:00 AM. What time will the 6th checkoccur?**

   Working: (6 − 1) × 13 = 65 hours = 2 days 17 hours. 7 AM + 17 hours = 12 AM.

   Answer: **12:00 AM (Monday, if the first check is Saturday)**

9. **Using a Caesar Cipher with a shift of 3, encrypt HELLO.**

   Working: H → K, E → H, L → O, L → O, O → R.

   Answer: **KHOOR**

10. **A factory machine is inspected every 18 days. The first inspection is on Thursday. On which weekday willthe 11th inspection occur?**

    Working: (11 − 1) × 18 = 180 days; 180 mod 7 = 5; Thursday + 5 = Tuesday.

    Answer: **Tuesday**

Real-Life Applications :

• Clock and calendar calculations

• Repeated-event scheduling

• Computer programming

• GPS and navigation

• Basic cryptography such as Caesar Cipher.

Result : The given clock, weekday, recurring-event and Caesar Cipher problems were successfully solved using modular arithmetic and remainder-based reasoning.

Conclusion : Modular arithmetic is useful for solving cyclic and repeating problems. It provides a simple method for calculations involving time, weekdays, schedules and introductory cryptographic logic.